

17.2 - Normal Curves and Normal Distributions

Problem Solving, Math 1000

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1 Definition

A **normal curve** is a perfect bell-shaped curve. It is the continuous version of the perfectly bell-shaped bar graph, and we can use the properties of this continuous version to analyze the properties of normal and approximately normal distributions.

Normal curves have several nice mathematical properties:

1. **Symmetry**: Every normal curve has a vertical axis of symmetry. That is, every normal curve may be split by a horizontal line to form two identical mirror images. We call this horizontal line the **axis of symmetry**.
2. **Median and mean**: We call the point of intersection of the horizontal data axis and the axis of symmetry the **center** of the distribution. The center corresponds to both the median, denoted M , and mean, denoted μ . That is, in a normal distribution, the median and the mean are equal (i.e. $M = \mu$). In an approximately normal distribution, the median and the mean are approximately equal (i.e. $M \approx \mu$).
3. **Standard Deviation**: The **standard deviation**, traditionally denoted by σ , is a measure of the "spread" of the data. In a normal distribution, this corresponds to a **point of inflection** in the normal curve – that is, the point where the curve transitions from going "up" to going "down."

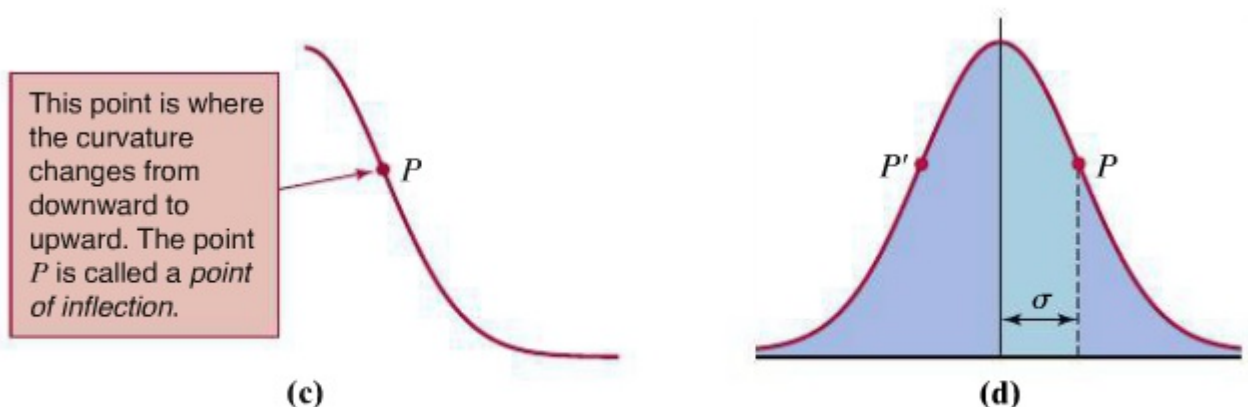


FIGURE 17-7

4. **Quartiles**: A data set may be divided into quartiles, which are "chunks" that partitioning the values of the data. The median separates the top half of the data from the

lower half, essentially partition the data into "low" and "high" sets. Going further, the point between the minimum and the median divides the lower half, while the point between the maximum and the median divides the top half. All together, this splits the data into four sections, denoted by the cut points Q_1 , Q_2 , and Q_3 . Each of these comprises a fourth of the data. In a normal distribution, $Q_3 \approx \mu + (0.675)\sigma$ and $Q_1 = \mu - (0.675)\sigma$.

5. **The 68 – 95 – 99.7 Rule:** When we look at a typical bell-shaped distribution, we may see that most of the data are concentrated near the center. As we move away from the center, the amount of data drops rather quickly. This formalizes in the **68 – 95 – 99.7 rule**, which says that approximately 68% of all the data falls within one standard deviation above or below the mean, 95% of all the data falls withing two standard deviations above or below the mean, and 99.7% of all the data falls within three standard deviations above or below the mean.

2 Example

the following graphs show a variety of normal curves, illustrating the notion of a "bell" shape to which we keep referencing.

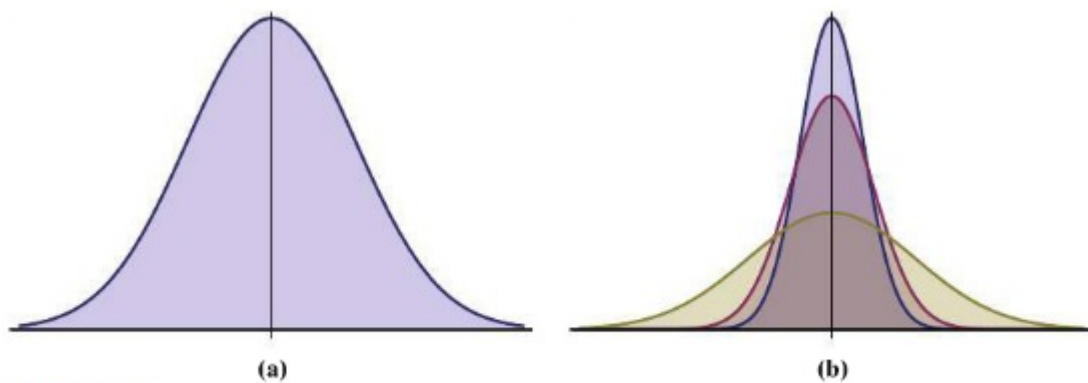


FIGURE 17-6 The many faces of normal curves.

3 Definition

The **range**, denoted R , of a data set is defied to be $R = Max - Min$, where Max is the maximum observed value and Min is the minimum observed value.

4 Question

Bell curves seem to come in different shapes, some short and squat, some tall and narrow. Why does this happen? Are these curves different in a particularly meaningful way?

5 Definition

We may re-frame all normal distributions to, essentially, "look" the same by using only the language of the mean, μ , and the standard deviation, σ , to talk about the data. This process is called **standardizing** the data.

6 Definition

When standardizing data, we rephrase our observations by use of a standardized data value called a **z-value**. A **z-value** is a measure of how far an observed value has strayed from the mean in terms of the standard deviation.

In a normal distribution with mean μ and standard deviation σ , the standardized data value of a data point, x , is given as a z-value by $z = (x - \mu)/\sigma$.

7 Example

Consider a normally distributed data set with mean $\mu = 63.18$ and standard deviation 13.27. Then the z-value of the observation $x = 91.54$ is given by

$$z = (x - \mu)/\sigma = (91.54 - 63.18)/13.27 \approx 2.14$$